Vectors

Section 1: 3D Vector Arithmetic & Magnitude

1 Given $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$, find the unit vector in the direction of $\mathbf{a}$.

a. $\frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$

b. $\frac{2}{49}\mathbf{i} - \frac{3}{49}\mathbf{j} + \frac{6}{49}\mathbf{k}$

c. $\frac{1}{7}(2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})$

d. $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$

2 Find the distance between the points $P(1, -2, 4)$ and $Q(4, 2, -8)$.

a. 13

b. 12

c. $\sqrt{153}$

d. 14

3 If $\mathbf{u} = [1, 2, -1]$ and $\mathbf{v} = [3, -1, 2]$, compute $2\mathbf{u} + 3\mathbf{v}$.

a. $[11, 1, 4]$

b. $[11, 7, 4]$

c. $[8, 1, 4]$

d. $[7, 3, 2]$

Section 2: The Scalar (Dot) Product & Angles

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4 Compute the scalar product $\mathbf{a} \cdot \mathbf{b}$ where $\mathbf{a} = [3, -2, 4]$ and $\mathbf{b} = [1, 5, 2]$.

a. 1

b. 5

c. -1

d. 15

5 For what value of k are the vectors $\mathbf{u} = [2, k, -3]$ and $\mathbf{v} = [4, 2, 6]$ perpendicular?

a. $k = 5$

b. $k = -5$

c. $k = 10$

d. $k = 0$

6 Find the acute angle between vectors $\mathbf{a} = [1, 1, 0]$ and $\mathbf{b} = [0, 1, 1]$.

a. 60°

b. 45°

c. 30°

d. 90°

Section 3: Vector Equations of Lines in 3D

7 Find the vector equation of the line passing through point $A(1, 2, 3)$ in the direction $\mathbf{d} = [2, -1, 4]$.

a. $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$

b. $\mathbf{r} = [2, -1, 4] + \lambda[1, 2, 3]$

c. $\mathbf{r} = [1, 2, 3] \cdot [2, -1, 4]$

d. $\mathbf{r} = [3, 1, 7] + \lambda[2, -1, 4]$

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8 Determine whether the point $(5, 0, 11)$ lies on the line $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$.

a. Yes, corresponding to $\lambda = 2$

b. No, λ values are inconsistent

c. Yes, corresponding to $\lambda = 3$

d. No, because the z-coordinate is too large

9 Find the point of intersection of the lines $\mathbf{r}_1 = [1, 0, 1] + \lambda[1, 2, -1]$ and $\mathbf{r}_2 = [0, 4, 3] + \mu[2, -1, -2]$.

a. $(2, 2, 0)$

b. $(1, 0, 1)$

c. $(3, 4, -1)$

d. The lines are skew

Section 4: Skew Lines & Shortest Distances

10 Two lines in 3D do not intersect and are not parallel. What term describes such lines?

a. Skew lines

b. Coplanar lines

c. Asymptotic lines

d. Coincident lines

11 Find the acute angle between the line $\mathbf{r} = [1, 0, 2] + \lambda[1, -2, 2]$ and the line $\mathbf{r}' = [3, -1, 4] + \mu[3, 0, 4]$.

a. $\arccos(11/15) \approx 42.8^\circ$

b. 30°

c. 45°

d. 60°

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12 Find the coordinates of the foot of the perpendicular from the origin to the line $\mathbf{r} = [1, -1, 2] + \lambda[2, 1, -1]$.

a. $[1/3, -4/3, 7/3]$

b. $[1, -1, 2]$

c. $[0, 0, 0]$

d. $[2, 1, -1]$

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